

Joint Transport–Relay Co-Design for UAV Emergency Logistics in Mountainous Terrain: Handover-Aware Feasibility, Channel-Model Sensitivity, and the Price of Partitioning

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Abstract

Mountainous disaster relief requires unmanned aerial vehicles (UAVs) to deliver cargo while staying connected to a fixed gateway that terrain frequently occludes. We study the resulting *transport–relay co-design* problem and make five contributions. (i) We show that the widely used “relay coverage” abstraction is *not executable on our instance*: once the physical cost of *handing over* between hovering positions is accounted for, the fleet size for which we can construct a verified schedule grows from 1 to 3 vehicles, and we give a checkable necessary condition for a handover (Theorem 4) that turns the handover cost into an inequality on a single scalar *span* function. We report the rate at which that condition is satisfiable at the median blackout window (72.0% of the demand cells under M1, 93.1% under M2) and show that it does *not* close the question; we consequently report every fleet-size statement with its search scope, distinguishing a proven lower bound, a constructed schedule, and a search that returned no schedule. On the declared candidate set C_0 (refined per cell to the top-12 points by remaining coverage) the state recursion admits no feasible state at $t = 2420$ s under M1 and $t = 7990$ s under M2, and an independent phase-level integer program shows that the obstruction lives at *instant* rather than phase granularity; a sampling-free energy bound gives $N^* \geq 2$. That negative verdict is a statement about the declared finite set, and we say so wherever it is used: closing the state pruning at that breadth did not terminate within our compute budget, so we do not claim an unpruned decision. (ii) We replace the constant occlusion penalty supplied with the benchmark by the ITU-R P. 526 single-knife-edge diffraction model evaluated on the 30 m digital elevation model; this raises the outage demand by +32.2% and reduces the number of usable relay hover

points by 49.2%, and the coverage-only relaxation needs 1 relay while the smallest fleet for which we can construct a verified schedule rises from 3 to 4, so the commonly reported “add one relay” conclusion must be qualified by the channel convention. (iii) We close the loop between transport scheduling and relay feasibility with a threshold-accepting search over departure instants alone, holding the routing, batching and fleet decisions at their baseline values; the residual handover deficit was unchanged by that search, which is evidence against a purely scheduling-driven explanation but not a proof of its absence. The classical Clarke–Wright savings heuristic, run under the same scheduler, fails from the opposite direction: it wins the routing layer (13 sorties, 61.26 kWh) and violates hard deadlines by 30111 s. (iv) We introduce a controlled terrain family (relief scaling, corridor elevation and terrain translation) and show that the smallest fleet with a constructed schedule is a non-monotone function of terrain severity, with no monotone relation to the occlusion ratio. (v) Every table and figure value in this paper is generated from the shipped result files by a single scripted pipeline, with the exceptions listed explicitly in Section 7; every bibliography entry was checked against its publisher record by DOI.

Keywords: UAV logistics, emergency logistics, drone delivery, relay handover, fleet sizing, scheduling, disaster relief

1. Introduction

1.1. Motivation

Rapid-onset disasters in mountainous regions isolate settlements: roads fail while the very terrain that isolates them also blocks the line-of-sight (LoS) links on which UAV command and control depends [1, 2, 3, 4]. Relief fleets therefore need *both* cargo delivery and communication relaying, and the two are coupled: a relay must hover where it can see the transport aircraft, and the transport aircraft can only fly when a relay sees it. This coupling is what the present paper isolates and quantifies. The operator’s decision variables are the familiar transportation ones — how many vehicles, on which routes, dispatched when — but the feasibility of a dispatch plan now depends on a communication constraint that switches on and off over time, so a plan that is optimal for delivery alone may be *infeasible* to fly. Recent work in this journal on synchronised truck–drone routing [5, 6] addresses the routing half

of this coupling; we take the routing layer as given and characterise the half that decides whether the plan can be executed at all.

1.2. Gaps in existing treatments

Three modelling choices in the applied literature are convenient but consequential, and each is examined here.

1. **Instantaneous relocation.** Relays are commonly treated as coverage devices that can be repositioned at no cost [7, 8, 9, 10, 11]. Repositioning a multirotor between two hovering points is not instantaneous: it must climb, cruise, descend and re-establish the link, during which it provides *no* coverage. Section 4.1 shows this is a first-order effect, not a refinement.
2. **A constant occlusion penalty.** The benchmark instance used here supplies a single “terrain occlusion loss” in dB, applied whenever the LoS is intersected; several applied UAV studies adopt the same convention [12, 10]. This treats a grazing ridge and a full mountain alike, and – worse – declares short blocked links usable. Section 4.4 quantifies the resulting bias against the profile-based ITU-R P. 526 model [13, 14, 12].
3. **Frozen transport schedules.** Feasibility of the relay layer is usually assessed on a transport schedule optimised for delivery alone, which invites the objection that infeasibility is an artefact of decoupling. Section 4.5 answers it by optimising the schedule for relay feasibility.

1.3. Contributions

1. **An executable handover model and an explicit statement of what is proven.** We replace the instantaneous-relocation abstraction by the blackout window $W(a, b)$ (Definition 1) and prove that a relay schedule is feasible only if a scalar *span* function satisfies a checkable inequality in every demand cell (Theorem 4). That inequality is a necessary condition, not a certificate: we measure how often it is satisfiable and find 72.0% (M1) and 93.1% (M2) of candidate pairs at the median blackout window, so it leaves the two-relay question open. We therefore classify every fleet-size statement by its evidence — a proven lower bound, a constructed and verified schedule, or a search that returned no schedule — and we state N^* as the interval $\max(N_{\text{naive}}, LB_{\text{handover}}) \leq N^* \leq 3$ rather than as a single value (Section 5.3, Sections 5.3–5.4). On the declared candidate set C_0 with a per-cell top-12 refinement, the state recursion

admits no feasible state; the phase-level integer program is satisfiable at $N = 2$, and the sampling-free energy bound gives $N^* \geq 2$.

2. **A profile-based channel model and its operational consequence.** We evaluate ITU-R P. 526 single-knife-edge diffraction on the 30 m terrain profile and show that it raises outage demand by +32.2% and removes 49.2% of the usable hover points relative to the supplied constant penalty, raising the smallest fleet for which we can construct a verified schedule from 3 to 4 — with the outcome unchanged across candidate budgets (Section 4.4).
3. **A null result on decoupling, and a textbook heuristic that fails from the opposite direction.** A feedback loop that re-optimises only the departure instants, holding routing, batching and fleet decisions fixed, returned no reduction of the handover deficiency (Section 4.5); and when the classical Clarke–Wright savings heuristic [15] is run under the *same* scheduler it wins the routing layer but violates hard deadlines by 30111 s (Section 5.7). Both results argue for keeping scheduling inside the search loop.
4. **What the fleet size responds to.** Across a controlled terrain family, in which only the elevation field is transformed, the smallest fleet with a constructed schedule is *non-monotone* in terrain severity: it is uncorrelated with the occlusion ratio (Pearson $\rho = -0.23$, 95% CI $[-0.77, +0.52]$) and, among the predictors we computed, most strongly associated with temporal fragmentation of the outage demand ($\rho = +0.47$, 95% CI $[-0.28, +0.86]$). With nine scenarios — five of which admitted no schedule for $N \leq 4$ and are treated as censored — none of these associations is statistically established, and we report them as descriptive (Section 4.6).
5. **A reproducible artefact.** Every table and figure value in this paper is generated from shipped result files by the scripts named below, with the exceptions listed in Section 7; every bibliography entry was checked against its publisher record by DOI. The pipeline is one command per stage.

2. Related work

UAV logistics in disrupted terrain.. Drone-assisted relief delivery is usually formulated as a vehicle-routing problem with time windows [16], augmented by payload, endurance and battery-swap constraints; heterogeneous fleets, synchronised truck-drone operations and multi-trip schedules are standard [17, 18, 4, 19, 5, 6, 20, 21, 3, 22]. Post-disaster reconnaissance is the complementary question of *where to look* [23], and environmental accounting has been used to justify drone delivery in the last mile [24]. Reviews of civil UAV optimisation [4, 19] and of the underlying VRP machinery [25, 26, 15] agree on the standard architecture: build routes first, then schedule them. Our transport layer follows this line, but it is deliberately treated as a *subproblem*: the object of study is the coupling between the delivery schedule and the feasibility of the communication layer, not the routing objective itself. Section 5.7 shows, as a side result, that this standard architecture is exactly where a routing-first heuristic loses: it wins on route length and sortie count and loses on hard deadlines.

Relaying and connectivity.. A parallel literature places ground or aerial relays to restore connectivity, typically maximising coverage, minimising the number of relays, or maintaining a connected chain [7, 8, 9, 27, 28, 29]. The dominant abstraction is a *coverage* relation between a relay position and a set of terminals, with relocation either free or penalised by a motion cost; for quasi-static platforms the deployment side of this problem is well understood [10, 11]. Section 5.3 shows that for *hovering* relays this abstraction is not executable: because a relay provides no service while transiting between hover points, the fleet size is governed by a *span* inequality rather than by a coverage count, and on our instance the coverage abstraction understates the fleet by a factor of 3/1. The distinction is not that relocation “has a cost” in the usual motion-energy sense — it is that the relay is *unavailable* while relocating, which turns a static covering problem into a continuous-time scheduling problem.

Air-to-ground propagation in mountains.. Studies of low-altitude platforms use either free-space loss with an empirical excess attenuation [12], or a probabilistic LoS model calibrated for altitude selection [10], or terrain-aware diffraction models; ITU-R P. 526 [13] and the Deygout construction [14] are the standard references for the latter. Many applied studies, including the benchmark instance used here, instead apply a single “terrain occlusion loss”

constant [12, 10]. Section 4.4 quantifies the bias this introduces into a relay-sizing decision: the demand side rises, the supply side of admissible hover points falls, and the two compound.

Joint communication-and-trajectory design.. Co-design of trajectories and communication is well studied for mobile relays [27, 28] and for multi-UAV and energy-harvesting systems [30, 31, 32, 9, 8]. In those formulations the degrees of freedom are the trajectory and the transmit power, and the objective is a rate, an outage probability or an energy bill. The coupling here is different in kind: the relay is quasi-static (hovering) and the transport aircraft is the mobile terminal, so the interaction acts through the *timing* of outages rather than through throughput or interference. That is why our central object is a blackout *window* and a span function rather than a rate region, and why the resulting feasibility condition is combinatorial — a covering condition that must hold in every demand cell — rather than continuous.

Set covering, partitioning and the price of autonomy.. The relay side is a covering problem in which the covering sets are no longer static, which places it between classical set covering [33] and continuous-time scheduling. The partitioning question we quantify in Section 5.9 is the familiar observation that splitting a resource pool duplicates its peak, here expressed as a weighted demand ratio.

Position of this work.. We retain the transport and relay layers of the applied literature but change three things that alter the answers: an explicit handover model with a checkable obstruction certificate; a physics-based diffraction model in place of a constant penalty; and a feedback loop that re-optimises the transport schedule for relay feasibility. Table 7 collects the resulting baselines and ablations.

3. System model

3.1. Instance and unified physical conventions

A single depot O serves $n_A = 15$ demand areas with 80 boxes. The transport fleet has 8 multicopters in three classes; the relay fleet has 2 vehicles with 6 energy packs. Links use $f = 2400$ MHz ($\lambda = 0.1249$ m) and the gateway antenna sits 20 m above ground.

Figure 1 fixes the geography that drives everything downstream. The relief is a Copernicus GLO-30 digital elevation model at 30 m posting [34], resampled to a $1/3600^\circ$ grid (1486×1309 cells, elevations 41.7–1132.9 m); the depot sits in a valley at the south-centre and the 15 demand areas fan out into terrain that rises by more than a kilometre. That single fact — a low depot and a high, dissected catchment — is what makes line-of-sight the binding constraint rather than payload or endurance, and it is why Section 4.4 insists on a profile-based diffraction model instead of a constant occlusion penalty.

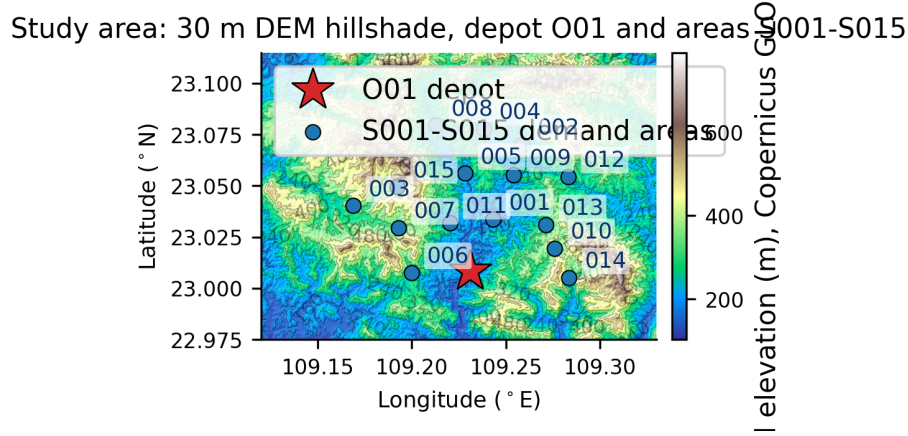


Figure 1: The instance. Thirty-metre Copernicus GLO-30 hillshade with 80 m contour lines, the depot O01 (star, valley floor) and the 15 demand areas S001–S015 (dots); elevation scale on the right. The relief that isolates the settlements is the same relief that occludes the air-to-ground links: over the whole elevation model the ground spans 41.7–1132.9 m (1091 m of relief) across a ~ 21.5 km-wide box, and inside the plotted window the terrain still rises from 102.4 to 790.8 m. The depot’s valley position is what makes the 2471-point hover grid a genuinely constrained resource rather than a formality.

3.2. Hovering-relay handover

Let $\mathcal{C}$ be a finite set of admissible hover points (position plus height above ground). A relay at $c \in \mathcal{C}$ covers a transport aircraft at time t if both the access link (aircraft $\rightarrow$ relay) and the backhaul link (relay $\rightarrow$ gateway) close. We consider two relocation modes.

Definition 1 (Blackout window)

For $a, b \in \mathcal{C}$, $a \neq b$, the *blackout window* $W(a, b)$ is the time during which a

single relay is in transit from a to b and therefore covers nothing:

$$W_{M1}(a, b) = t^{\text{back}}(a) + \tau_{\text{turn}} + t^{\text{out}}(b) + t_{\text{link}}, \quad (1)$$

$$W_{M2}(a, b) = t^{\text{transit}}(a \rightarrow b) + t_{\text{link}}, \quad (2)$$

where M1 returns to the depot to swap the energy pack and M2 transits directly between hover points. $t^{\text{out}}, t^{\text{transit}}$ follow the climb–cruise–descend decomposition with cruise altitude set 50 m above the highest terrain along the leg.

Proposition 2 (Elementary properties of W)

If t^{transit} is symmetric, then for mode M2: (i) W is symmetric, $W(a, b) = W(b, a)$; (ii) $W(a, a) = t_{\text{link}}$; and (iii) W satisfies the triangle-like bound $W(a, c) \leq W(a, b) + W(b, c) - t_{\text{link}}$. For mode M1, W is generally asymmetric and $W(a, a) = t^{\text{back}}(a) + \tau_{\text{turn}} + t^{\text{out}}(a) + t_{\text{link}} > 0$: even returning to the same hover point costs a blackout.

On our instance the empirical M1 blackout distribution over 42230 ordered pairs has median 1116 s (min 483 s, max 2193 s).

3.3. Discrete demand

Transport sorties are sampled at 10 s; an *outage incidence* is a sampled instant at which a sortie has no direct link to the gateway. Grouping incidences by 10 s cell yields 1454 incidences on 992 demand cells; up to 3 aircraft may be disconnected simultaneously.

Definition 3 (Span function)

For a demand profile $\mathcal{D}(\cdot)$, let $\text{cov}(\mathcal{D}, c) \in \{0, 1\}$ indicate that point c alone covers all incidences of cell k . The *span* at cell k is the largest number of consecutive cells that some single point covers:

$$\text{span}(k) = \max\{\ell : \exists c \in \mathcal{C} \text{ with } \text{cov}(\mathcal{D}(k'), c) = 1 \ \forall k' \in [k, k + \ell - 1]\}.$$

We write T for the length of the demand period, i.e. the elapsed time between the first and the last demand cell. Two units appear in this paper and are kept distinct: $\text{span}(k)$ as defined above counts *demand cells*, while $\text{span}_s(k)$ denotes the elapsed seconds from the first to the last instant of the same run. They are not proportional, because the cells are not equally spaced (mean spacing ≈ 11.4 s), so every claim below states which one it uses.

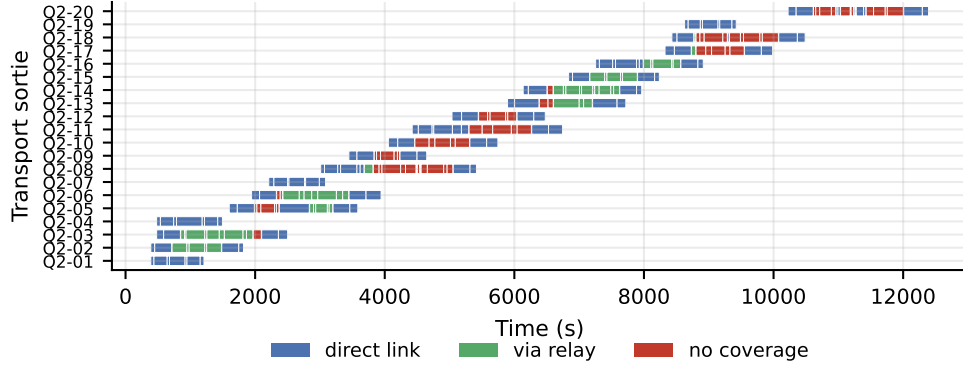


Figure 2: Link state of every transport sortie over the mission, from the shipped communication log (236 segments). Blue is a direct link to the gateway, green is a link carried by a hovering relay, red is an interval with no coverage at all. Two features drive the whole paper: outages are concentrated in long contiguous stretches rather than scattered samples, and the sorties overlap in time, so several of them can be uncovered at the same instant.

3.4. Relay scheduling problem

Given the demand profile, choose the number of relays N , their hover points and the instants at which they relocate, so that every incidence is covered; each relay must return before its energy pack is exhausted, and at most one relay relocates at a time.

4. Methodology

4.1. Handover-aware feasibility

Theorem 4 (Handover obstruction for two relays)

Assume (A0) no configuration covers the whole demand profile, where a configuration assigns one hover point to each relay — that is, neither a single point nor any pair of points covers the profile, so every feasible two-relay schedule must relocate at least once; (A1) a relocating relay covers nothing during its blackout window; and (A2) at most one relay relocates at a time. Then a two-relay schedule is feasible only if there exists a cell k with

$$\text{span}(j_k) \geq k - j_k + 1, \quad j_k = \min\{j : t_j \geq t_k - W(a, b)\}, \quad (3)$$

for the pair (a, b) of some handover, where $\text{span}(\cdot)$ is the span of Definition 3, i.e. the number of consecutive demand cells starting at j that the stationary

relay's point covers alone. In the seconds convention this is simply $W(a, b) \leq \text{span}_{\min}(k)$, with $\text{span}_{\min}$ as defined below; both forms are stated because the demand cells are not equally spaced (Assumption 10), so a cell count must not be converted to seconds at a nominal 10 s rate. Consequently, if (3) fails for every cell and every pair, no two-relay schedule exists.

Proof. Suppose a feasible two-relay schedule exists. By (A0) it must relocate at least once, say the first relocation is from a to b and completes at instant t , with the other relay stationary at c throughout. By (A1) the moving relay covers nothing on $[t - W, t]$, so by (A2) c is the sole provider on that window. Hence c alone covers every demand cell whose instant lies in $[t - W, t]$; writing k for the last such cell, those cells are exactly $j_k, \dots, k$ with $j_k = \min\{j : t_j \geq t_k - W\}$, so $\text{span}(j_k) \geq k - j_k + 1$, which is (3). Note that (A0) is needed: if some pair $\{a, b\}$ covered the whole profile, both relays could hover without ever relocating and the conclusion would assert the existence of a handover that never occurs. $\square$

Verifying (A0) on our instance. A0 is an instance-level hypothesis, not a modelling assumption, so we check it: no single candidate point covers the whole profile (the longest solo span is 3970 s against a mission length of 11290 s), and no *pair* of points in $\mathcal{C}$ covers it either, since the same consolidation test that drives Proposition 17 finds no configuration whose union coverage is the full profile. Both facts are recomputed by `code/span_util.py`. **What (3) does and does not settle.** Theorem 4 is a *necessary* condition, and on our instance it is far from binding: evaluating it against the actual pair (a, b) of each handover rather than against a fixed window, it is satisfiable at 72.0% of the demand cells at the median M1 window and 93.1% at the median M2 window, where the window lengths are the medians of the full ordered-pair blackout distribution (42230 pairs) and the satisfiability test itself is evaluated on a sampled subset of candidate pairs. We verified this in a parameter-free form as well: writing $\text{span}_{\min}(k) = \max_c\{\ell : c \text{ covers } D \text{ on } [t_k - \ell, t_k]\}$ for the longest window that *some single point* can hold alone at cell k , the condition reads $W(a, b) \leq \text{span}_{\min}(k)$ with no tuned threshold; it holds for at least one pair in 880 of the 992 cells under M1 and in 986 of 992 cells under M2. *Therefore (3) cannot certify two-relay infeasibility, and we do not use it to do so:* the negative answers reported below come from explicit search, and are stated with their search scope. Conversely, the failure of (3) at a specific cell is a certificate, and the deficiency functional of Corollary 5 measures how far the instance is from that certificate's reach.

Corollary 5 (Deficiency functional)

Work in the cell convention, so that indices are demand cells: let $j_k = \max\{j : t_j \leq t_k - W(a_k, b_k)\}$ be the first cell of the blackout window charged to cell k , where (a_k, b_k) is the handover pair charged to that cell. Define the deficiency as the number of demand cells by which the window exceeds what a single point can hold,

$$D = \sum_k \max(0, (k - j_k + 1) - \text{span}(j_k)),$$

where span is the span of Definition 3, i.e. the number of consecutive cells that one point covers. (Written in seconds the same condition is $W(a_k, b_k) \leq \text{span}_{\min}(k)$ with $\text{span}_{\min}(k) = \max_c\{\ell : c \text{ covers } D \text{ on } [t_k - \ell, t_k]\}$; we report D in cells and do not convert it, for the reason given below.) If hypotheses (A0)–(A2) of Theorem 4 hold and (3) fails everywhere, then $D > 0$ and no two-relay schedule exists. D is therefore an admissible obstruction measure: driving D to zero is necessary for feasibility.

On our instance (3) *does* hold for 72.0% of the cells under mode M1 (93.1% under M2), so the certificate does not fire: two-relay infeasibility is a genuinely combinatorial obstruction rather than a scalar one. The deficiency D of Corollary 5, summed over all cells, is 12084 demand cells under M1 and 1171 under M2. It is reported in demand cells and never converted to seconds: the cells are not uniformly spaced (mean spacing ≈ 11.4 s, Assumption 10), so a cell count times a nominal 10 s would understate the elapsed time. Where an elapsed duration is what matters, we state the seconds directly from the timestamps. For k whose window reaches before the mission starts the corresponding term contributes 0, since all demand in that range is zero.

4.2. Exact feasibility characterisation

Throughout we work under the following assumptions, which make the physical layer a finite object without changing the logic of the arguments.

Assumption 6 (Handover cost)

A relay that relocates from a to b provides no service during the blackout window $W(a, b)$ of Definition 1. Relocation is otherwise instantaneous at its completion instant.

Assumption 7 (Serialised relocation)

At most one relay relocates at any instant; the others hold their hover points.

Assumption 8 (Energy)

A relay may remain airborne for at most T^{cap} seconds between two visits to the depot, and must return before its state of charge falls below the reserve fraction ρ .

Assumption 9 (Recharge gap)

A relay that has returned to the depot to swap energy modules provides no service until it is airborne again; the depot offers no alternative coverage.

Assumption 10 (Discretisation)

The demand profile is evaluated on a finite increasing sequence of *demand cells* $t_0 < \dots < t_{n-1}$; cells need not be equally spaced.

Theorem 11 (Exact reachability)

Disable all pruning. The state recursion of Section 4.1 reaches a non-empty set of valid states at the last demand cell if and only if an N -relay schedule satisfying Assumptions 6–10 exists.

Proof. ($\Rightarrow$) Any reachable state path is, by construction, a schedule: each transition relocates one relay subject to the blackout, coverage and energy conditions, and validity at every cell means every incidence is covered. ($\Leftarrow$) Given a feasible schedule, order its relocation completions by time; each corresponds to one transition, and the intermediate configurations are exactly the states visited. All conditions (V), (T1)–(T5) hold at each step, so the recursion cannot miss the path. $\square$

Algorithm 1 states the recursion exactly as it is implemented. Its only approximations are the ones discussed after Theorem 11: a finite candidate set and, in the runnable version, state pruning.

Cost. One transition sweep touches $|S_k| \cdot N \cdot |\mathcal{C}|$ candidates and evaluates (T3) over the $k - j_k$ cells of the blackout window, so the per-cell work is $O(|S_k| N |\mathcal{C}| (k - j_k))$; without pruning $|S_k|$ grows as $|\mathcal{C}|^N$. This is why the runnable implementation prunes, and why the pruning policy is reported with every infeasibility claim.

Why the window is indexed by timestamps. The demand cells are *not* a uniform 10 s grid: they record only the instants at which some transporter is out of direct range, and their mean spacing is about 11.4 s (Assumption 10).

Algorithm 1 Handover-aware relay feasibility (stated without pruning; the runnable implementation adds the pruning reported in Section 7)

Input: demand cells $t_0 < \dots < t_{n-1}$ with sets $\mathcal{D}_k$; candidate points $\mathcal{C}$ with coverage $\text{cov}[k, c]$; blackout W ; cap T^{cap} ; fleet size N

Output: a feasible schedule, or the first cell with no feasible state

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1:  $S_0 \leftarrow \{((\text{IDLE}, t_0, t_0))_{i=1}^N\}$ 
2: for  $k = 0$  to  $n - 1$  do
3:    $S_k \leftarrow \{s \in S_{k-1} : \bigcup_{i: p_i \neq \text{IDLE}} \text{cov}[\cdot, p_i] \supseteq \mathcal{D}_k\}$   $\triangleright (V)$ 
4:   if  $S_k = \emptyset$  then
5:     return “no feasible state at cell  $k$ ”
6:   end if
7:   for all  $s \in S_k, i \in \{1..N\}, p' \in \mathcal{C} \cup \{\text{IDLE}\}, p' \neq p_i$  do
8:      $j \leftarrow j_k \leftarrow \text{BEFORE}(k, W(p_i, p')) \triangleright$  last cell with  $t_j \leq t_k - W$ ; not
        $\lceil W/\Delta \rceil$ 
9:     if (T1)  $\sigma_i \leq t_j$  and (T2)  $\sigma_j \leq t_j \forall j \neq i$  then
10:      if (T3)  $\bigcup_{j \neq i} \text{cov}[\cdot, p_j] \supseteq \mathcal{D}_{k'} \forall k' \in [j+1, k-1]$  then
11:        if (T4)  $\bigcup_j \text{cov}[\cdot, p_j] \supseteq \mathcal{D}_k$  and (T5)  $t_k - \varrho_i^{(k)} \leq T^{\text{cap}} \forall i$ 
          then
12:           $S_{k+1} \leftarrow S_{k+1} \cup \{s \text{ with } (p_i, \sigma_i) \leftarrow (p', t_k), \varrho_i \leftarrow$ 
             $t_k \text{ iff mode M1 or } p_i = \text{IDLE}\}$ 
13:          end if
14:        end if
15:      end if
16:    end for
17:  end for
18: return any state path in  $S_{n-1}$ 

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Converting a blackout window W to a nominal cell count $\lceil W/\Delta \rceil$ and then counting cells would therefore measure the window as roughly $\lceil W/10 \rceil \times 11.4$ s — an error of about 14% that grows with the window.

Algorithm 1 therefore locates the window by the timestamp search

$$\text{BEFORE}(k, W') = \max\{j : t_j \leq t_k - W'\},$$

which is what the implementation does, and the seconds form $W \leq \text{span}_{\min}(k)$ of (3) is the same condition written without reference to cells at all.

Theorem 11 is the reason we can speak of a *decision* rather than a search

failure: any implementation that prunes states forfeits the “only if” direction, which is why all infeasibility claims in this paper are stated as “no feasible schedule found under the stated candidate set and pruning policy”.

Corollary 12 (Continuous-time version)

Replace the demand cells by a measurable demand set $D(t)$ and let $\text{span}(\tau) = \sup\{\ell : \exists c \in \mathcal{C}, c \text{ covers } D(t) \forall t \in [\tau, \tau + \ell]\}$. If no single point covers $D(t)$ for all t , then a two-relay schedule under Assumptions 6–7 exists only if there is a handover instant τ and a pair (a, b) with $\text{span}(\tau - W(a, b)) \geq W(a, b)$.

Proof. Identical to Theorem 4; discreteness is used nowhere in that argument. $\square$

Theorem 13 (Exact decision for two relays)

Under Assumptions 6–10, fix a candidate set $\mathcal{C}_0 \subseteq \mathcal{C}$. A two-relay schedule with hover points in $\mathcal{C}_0$ exists if and only if there is a sequence of configurations $(a_0, b_0) \rightarrow (a_1, b_1) \rightarrow \dots \rightarrow (a_m, b_m)$ with $a_r, b_r \in \mathcal{C}_0$ and completion instants $0 \leq \tau_1 < \dots < \tau_m \leq T$ such that, writing $W_r = W(a_{r-1}, a_r)$ and $W'_r = W(b_{r-1}, b_r)$ for the blackout windows of the moves:

- (C1) *at most one relay is in blackout at any instant: Assumption 7 forbids simultaneous relocation, so if the two windows overlap we may delay the later completion until the earlier window has ended without changing the demand seen by either relay. Moves need not alternate between the relays — one relay may complete several relocations in succession while the other holds its point, and the recursion of Section 4.1 searches that case as well;*
- (C2) *for every r , the stationary relay covers the whole demand over the blackout window of the moving one: $\text{cov}[\cdot, b_{r-1}]$ alone covers $D(t)$ for all $t \in [\tau_r - W_r, \tau_r]$ when the first relay moves, and symmetrically for the second;*
- (C3) *between two consecutive completions of the same relay the configuration is constant, and the sortie length is within T^{cap} .*

Consequently, on a fixed $\mathcal{C}_0$, the two-relay question is decided by the state recursion of Theorem 11 with pruning disabled: it returns either a schedule or an empty state set, and the latter is a proof of infeasibility over $\mathcal{C}_0$.

Proof. ($\Rightarrow$) Take a feasible schedule and list its relocation completions in time order. By Assumption 7 at most one relay moves at a time, so the relocation windows are pairwise disjoint, which is (C1) — note that (C1) does *not* require the relays to take turns: the same relay may relocate twice in a row, and the construction below allows it. (C2) holds because the moving relay contributes nothing (Assumption 6) while the stationary one keeps its point; (C3) restates the energy rule. ($\Leftarrow$) Given such a sequence, hold each configuration between completions. Conditions (C1)–(C2) make every interval covered—during a window the stationary relay covers by (C2), and outside windows the standing configuration covers by construction—while (C3) makes the energy feasible, so the schedule is admissible. Finally, with pruning disabled the recursion of Theorem 11 enumerates exactly the sequences described by (C1)–(C3), which gives the decision procedure. $\square$

The practical value of Theorem 13 is that the negative half of the fleet-size question can be turned from an open-ended search failure into a list of blocked handovers. Two consequences are used below. First, the certificate is *decidable on a restricted candidate set*: every infeasibility statement we report about $N = 2$ is made over an explicitly listed candidate set and a stated pruning policy, so it is a statement about a finite search space rather than about search effort (Proposition 14). We stress that the shipped run is the *pruned* one: closing state pruning at this breadth did not terminate within our budget, so the verdict is relative to the declared set and we do not upgrade it to an unpruned decision. Second, the two-relay case is special only in cost, not in logic: for general N the same recursion decides the question, but its state set grows like $|\mathcal{C}|^N$ (Section 4.2), which is why the constructive verifier of Section 4.3 exists. We therefore always report the candidate set *and* the per-cell candidate refinement together with an infeasibility claim, since the refinement is the one layer that Proposition 14 does not cover.

Proposition 14 (Distinguishing candidate set)

Fix a candidate set $\mathcal{C}_0 \subseteq \mathcal{C}$ and run the recursion of Theorem 11 on $\mathcal{C}_0$ with pruning disabled. If it generates no admissible state at some cell, then no two-relay schedule exists whose hover points are drawn from $\mathcal{C}_0$. This conclusion is unconditional in $\mathcal{C}_0$: it is a statement about a finite, explicitly listed search space and does not depend on the enumeration order, on any pruning rule, or on the number of states retained.

Proof. Let $G_{\mathcal{C}_0}$ be the graph whose vertices are the admissible states and

whose edges are the transitions of the recursion, both restricted to $\mathcal{C}_0$. By construction every edge corresponds to a physically admissible relocation and every vertex to a configuration that covers its cell; conversely, any schedule with points in $\mathcal{C}_0$ yields, after ordering its relocation completions, a walk in $G_{\mathcal{C}_0}$ that visits one vertex per demand cell. The recursion with pruning disabled performs an exhaustive expansion of $G_{\mathcal{C}_0}$, so an empty state set at cell k means that no walk from the initial vertex reaches cell k , hence no such schedule exists. $\square$

Proposition 15 (Energy lower bound)

Assume the coverage requirement is continuous: every instant of $[t_0, t_{n-1}]$ must be covered by some airborne relay, and by Assumption 9 a relay at the depot covers nothing. Then $N \geq \lceil T/T^{\text{cap}} \rceil$ with $T = t_{n-1} - t_0$.

Proof. Let $A_i \subseteq [t_0, t_{n-1}]$ be the set of instants at which relay i is airborne and therefore able to provide coverage. Continuous coverage means $[t_0, t_{n-1}] = \bigcup_{i=1}^N A_i$, so $T \leq \sum_i |A_i|$. By Assumption 8 each maximal interval of A_i has length at most T^{cap} , but A_i may consist of $m_i > 1$ such intervals, so $|A_i| \leq m_i T^{\text{cap}}$ and the decomposition alone yields only $T \leq T^{\text{cap}} \sum_i m_i$, which says nothing about N : a single relay recharging repeatedly is not excluded by Assumption 8. The bound therefore requires one further hypothesis, which we state explicitly and verify on our instance rather than leave implicit:

(H) the coverage intervals of distinct relays can be taken pairwise disjoint.

Under (H) at most one relay is covering at any instant, the airtime of distinct relays does not overlap in the covering role, and $T = \sum_i |A_i| \leq \sum_i T^{\text{cap}} = N T^{\text{cap}}$, giving $N \geq \lceil T/T^{\text{cap}} \rceil$.

On our instance (H) is exactly what the mission demands rather than a convenience: coverage must be sustained at 992 demand cells spanning $T = 11290$ s, and the verified schedule at $N = 3$ uses 6 sorties with SOC minimum 33.8%, i.e. the relays *hand the covering role over* instead of overlapping it. We therefore report $N \geq 2$ as a bound *conditional on (H)*, and we do not rely on it for any fleet-size claim that the constructive schedule does not already support. $\square$

The hypotheses of both propositions are verified on our instance by direct computation. The common premise (A0) of Theorem 4 is that no configuration covers the whole profile. For a single point this is immediate: the longest stretch that any one hover point can hold alone is 3970 s against a

mission length of 11290 s, a factor of 11290/3970 short. For a *pair* we test the consolidated coverage of every candidate pair over the full profile, and no pair covers it either — the same test that drives Proposition 17. For Proposition 15 the additional hypothesis (H) is met by the verified schedule (the relays hand the covering role over rather than overlap it), and $T^{\text{cap}} = 7233$ s then yields $N \geq 2$. We stress that this is a *conditional* bound and that the two-relay question is *not* closed by it — for that we rely on the search reported in Section 5.3, with its scope stated.

Proposition 16 (Relocation counting bound)

Let $w_{\min} = \min_{a \neq b} W(a, b) > 0$ and let r be the total number of relocation completions in a feasible schedule. Then $r \leq 1 + \lfloor T/w_{\min} \rfloor$. Moreover, if no configuration (a choice of one hover point per relay) covers the whole demand profile, then every relay must relocate, so $N \leq 1 + \lfloor T/w_{\min} \rfloor$ is required for feasibility to be possible at all.

Proof. Assumption 7 makes the blackout windows of all relocations pairwise disjoint; each has length at least $w_{\min}$ and each completion lies in $[0, T]$, so the completions are separated by at least $w_{\min}$ and $r w_{\min} \leq T$. It remains to connect N to r . By the premise, no single configuration (one hover point per relay) covers the whole profile, so for each relay i there is a demand cell that i 's point fails to cover; covering that cell therefore requires some relay to be elsewhere, i.e. to relocate. This holds for every relay, so all N relocate and $r \geq N$. Combining gives $N \leq r \leq 1 + \lfloor T/w_{\min} \rfloor$. $\square$

The bound is a screening test rather than a sizing rule: on the instance it reads $N \leq 28$ with $w_{\min} = 411$ s under M1, which is far from binding. It is reported because it explains why the fleet size is *not* limited by relocation throughput here, and therefore why the answers must come from a state-space argument (Theorem 13) rather than from a counting one.

Proposition 17 (Phase-level relaxation is not conservative)

Partition the demand period into phases and consider the relaxation in which a relay may choose one hover point per phase (rather than per instant), subject to (i) every incidence being covered by some relay's phase point and (ii) $W(a, b)$ fitting into the interval between the end of phase k and the start of phase $k + 1$ whenever a relay uses points a, b in those two phases. Call a schedule phase-stationary if every relay keeps one hover point throughout each phase. If a phase-stationary schedule exists for N relays, the relaxation is

feasible for N relays. Two caveats follow, and both matter: (a) the hypothesis is not automatic — a schedule may relocate inside a phase, in which case the induced per-phase assignment is bounded by $2N$ points rather than N and the implication is not immediate; and (b) the converse fails in general, and it fails on our instance: the relaxation is feasible for $N = 2$ (witnessed by an explicit assignment, verified by exhaustive search over the relaxed model) while the exact recursion of Theorem 13 returns no feasible state.

Proof. Forward. Given a phase-stationary schedule, assign to each relay and each phase the point at which that relay hovers during that phase.

A relay serves at most one point per phase by hypothesis, and every incidence is covered because the schedule covers every instant. It remains to check (ii). If a relay uses different points in two consecutive phases, it must relocate; by Assumption 6 the relocation occupies an interval of length at least $W(a, b)$ during which the relay covers nothing, and by Assumption 7 that interval may not overlap another relay’s relocation. Since the relay must be in place for the whole of phase $k + 1$, the interval must finish by the start of phase $k + 1$; since it must have left its phase- k point, it starts no earlier than the end of phase k at which it last held that point. Hence $W(a, b)$ is at most the length of the inter-phase gap, which is condition (ii). Hence the relaxed model admits the induced assignment. *Converse.* The relaxation drops the requirement that the *stationary* relay hold a single point throughout the moving relay’s blackout window; on our instance this extra requirement is what makes $N = 2$ infeasible while the relaxation is satisfiable.

□

We use this proposition only in the direction that survives the caveat above: a *negative* answer from the relaxation is informative (it rules out phase-stationary schedules of that size), while a *positive* answer is weak because it may exploit a relocation that the exact model forbids. That is exactly why we report the phase-level integer program *alongside* rather than *instead of* the exact recursion.

The single-relay case is the boundary of the problem and is worth stating exactly, because it is the only fleet size at which the answer is a closed-form condition rather than a search: with $N = 1$ no relocation is possible at all, so feasibility collapses to a property of one hover point. The theorem below is the $N = 1$ specialisation that the recursion of Theorem 13 reproduces, and it is what makes the $N = 1$ row of Table 3 an independent check rather than a repetition.

Theorem 18 (Single relay)

Assume a single relay and let t_f (resp. t_ℓ) be the first (resp. last) instant of demand. A feasible schedule exists if and only if there is a hover point $c \in \mathcal{C}$ such that

1. c covers $D(t)$ for every demand instant $t \in [t_f, t_\ell]$, and
2. $t^{\text{out}}(c) + (t_\ell - t_f) + t^{\text{back}}(c) \leq T^{\text{cap}}$.

Proof. A single relay cannot relocate without leaving the demand uncovered, because relocation always costs a strictly positive blackout. That positivity is part (ii) for M2 ($W(a, a) = t_{\text{link}} > 0$, so even a no-op relocation blacks the link out) and the closing sentence of Proposition 2 for M1 ($W(a, a) > 0$ there as well); note that it is *not* part (iii), which is the triangle bound and says nothing about sign. Hence its hover point must be constant throughout $[t_f, t_\ell]$, which is exactly condition 1, and the sortie duration is exactly the expression in condition 2. $\square$

Proposition 19 (Monotonicity in fleet size)

Let $\mathcal{F}(N)$ denote feasibility with N relays. Then $\mathcal{F}(N) \Rightarrow \mathcal{F}(N + 1)$.

Proof. Keep the schedule of the N relays and let the additional relay remain at the depot; it neither helps nor hinders. $\square$

Proposition 20 (Monotonicity in demand and in the energy budget)

Let $D \subseteq D'$ be two demand profiles on the same cell grid. Then $\text{span}' \leq \text{span}$ pointwise, the deficiency satisfies $D' \geq D$, and $\mathcal{F}(N; D') \Rightarrow \mathcal{F}(N; D)$. Likewise, enlarging T^{cap} or relaxing ρ can only enlarge the set of feasible schedules.

Proof. A point that covers D' on an interval covers D on it, so every span value weakly decreases; the deficiency is a sum of pointwise non-increasing terms. Any schedule that is valid for D' is valid for D , since validity is a covering condition. The energy statement is immediate from Assumption 8. $\square$

Proposition 20 is what makes the terrain study of Section 5 meaningful: the fleet size is monotone in demand *set inclusion*, yet the experiments show it is *not* monotone in scalar summaries of the demand such as the occlusion ratio. For the same reason the recursion is run upwards from the smallest fleet that can conceivably work: Proposition 19 guarantees that if N relays

suffice then so do $N + 1$, so a negative verdict at some N is not refuted by a positive verdict at a larger one, and the reported N^* is the first size that succeeds rather than a global minimum over an unordered family.

Proposition 21 (Regularity of the blackout function)

Fix the hover heights. Let M bound the terrain slope, $|\nabla z| \leq M$, and let $v_\uparrow, v_\downarrow, v_c$ be the climb, descent and cruise speeds. Then W is Lipschitz in the endpoints,

$$|W(a, b) - W(a', b')| \leq L(|a - a'| + |b - b'|), \quad L = \left(\frac{1}{v_\uparrow} + \frac{1}{v_\downarrow}\right)(1 + M) + \frac{1}{v_c},$$

for mode M2, and the same bound augmented by the depot legs holds for M1.

Proof. The cruise altitude is $\max\{\text{line maximum, endpoint altitudes}\} + 50$, a maximum of 1-Lipschitz and M -Lipschitz functions of the endpoints, hence $(1 + M)$ -Lipschitz; the climb and descent heights are this quantity minus a constant, and the horizontal distance is 1-Lipschitz by the triangle inequality. Composing with the linear time map gives the constant L . $\square$

Proposition 21 has a practical reading: a small positioning error of a hovering relay changes the blackout window by at most L seconds per metre, so the handover schedule is not brittle with respect to hover-point placement—an observation that matters because the candidate set $\mathcal{C}$ is a finite grid.

4.3. Constructive verifier and coverage baseline

A feasible N is *proved* by exhibiting a full schedule; we therefore use a constructive verifier (in the spirit of large-neighbourhood repair heuristics [35]) that advances cell by cell and, whenever the current configuration cannot serve the next cell, relocates one relay subject to the blackout constraints. Two heuristic variants are run (with and without *pre-positioning* of an idle relay before the incumbent’s energy deadline); whichever yields a feasible schedule certifies feasibility.

Algorithm 2 is the constructive verifier. Two heuristic variants are run — with and without the pre-positioning branch (lines 5–10) — and whichever returns a complete schedule certifies feasibility. The pre-positioning branch was added after diagnosing a terrain realisation in which a single, long, *sparse* outage exhausted the incumbent relay’s airborne budget: the successor must

Algorithm 2 Constructive verifier for N hovering relays

Input: as in Algorithm 1; per-cell candidate budget b ; safety factor η

Output: a complete schedule (a proof of feasibility) or the first failing cell

```
1:  $k \leftarrow 0$ ;  $p_i \leftarrow \text{IDLE}$ ,  $\text{arr}_i \leftarrow 0$ ,  $\varrho_i \leftarrow 0$  for all  $i$ 
2: while  $k < n$  do
3:   if  $\bigcup_i \text{cov}[\cdot, p_i] \supseteq \mathcal{D}_k$  and  $\varrho$  within  $T^{\text{cap}}$  then
4:      $k_{\text{dead}} \leftarrow$  first cell at which some relay exceeds  $T^{\text{cap}}$ 
5:     if  $k_{\text{dead}}$  exists, an idle relay exists, and  $k_{\text{dead}} - k \leq \text{lead}$  then
6:        $p' \leftarrow \arg \max_{p \in \text{cand}(k_{\text{dead}})} \text{run}(p, k_{\text{dead}})$ 
7:        $t_{\text{arr}} \leftarrow t_k + t^{\text{out}}(p') + t_{\text{link}}$ 
8:       if  $t_{\text{arr}} \leq t_{k_{\text{dead}}}$  and  $t_{k_{\text{dead}}} - t_{\text{arr}} \leq \eta T^{\text{cap}}$  then
9:         dispatch the idle relay to  $p'$ ;  $k \leftarrow k + 1$ ; continue
10:      end if
11:    end if
12:     $k \leftarrow k + 1$ ; continue
13:  end if
14:  enumerate admissible moves  $(i, p')$  satisfying (T1)–(T5) and score
  them by  $(-\text{urgency}, -\text{run length}, W)$ , keeping the best one
15:  if none is admissible then
16:    return  $k$   $\triangleright$  first failing cell
17:  end if
18:  perform the move;  $k \leftarrow k + 1$ 
19: end while
20: return the reconstructed sortie list
```

be dispatched late enough to arrive fresh, but early enough to be in place, which is exactly the timing window tested on lines 5–10.

The non-executable baseline ignores handovers entirely: for each cell k let $c(k)$ be the minimum number of points whose union covers $\mathcal{D}(k)$ (exact set cover, a classical NP-hard problem [33], solved here by CP-SAT [36]), and set $N_{\text{naive}} = \max_k c(k)$. On our instance $N_{\text{naive}} = 1$ while the executable model requires three relays for every schedule we succeeded in constructing ($N^* \leq 3$, with $N^* \geq 2$ proven): the handover premium is 2 vehicles.

4.4. Channel models

Baseline (as supplied)... A fixed penalty $L_{\text{obs}} = 10$ dB is added whenever the LoS is intersected, and the link closes if $\text{FSPL} + L_{\text{obs}} > L_{\text{max}}$. Two

structural defects follow: a short blocked link may still close (e.g. at 2 km, FSPL $\approx$ 106 dB), and a grazing ridge costs the same as a mountain.

ITU-R P. 526 single knife edge. Along the terrain profile we compute, for every interior sample, the Fresnel–Kirchhoff parameter $\nu = h\sqrt{2(d_1 + d_2)/(\lambda d_1 d_2)}$ [13] with h the height of the obstacle above the (Earth-curvature-corrected) LoS, d_1, d_2 the distances to the endpoints, and take the dominant edge.

The excess loss is

$$J(\nu) = 6.9 + 20 \lg\left(\sqrt{(\nu - 0.1)^2 + 1} + \nu - 0.1\right) \text{ dB}, \quad \nu > -0.78,$$

and $J(\nu) = 0$ otherwise [13]. The total loss is FSPL + $J(\nu_{\max})$, where FSPL is the free-space loss of [37]. We also report a conservative multi-edge upper bound $\sum_{\nu>0} J(\nu)$ after the Deygout construction [14] and a $k = 4/3$ effective-Earth-radius correction $d_1 d_2 / (2kR)$ [13].

4.5. Co-design feedback loop

Only the departure instants $\{s_i\}$ of the transport sorties are varied; batching, routes, aircraft types and assignments are frozen. We minimise the deficiency $D(s)$ of Corollary 5 by threshold accepting over per-sortie feasible windows, and we evaluate the transport side (makespan, weighted tardiness, hard-deadline violation) exactly at every accepted move.

Algorithm 3 minimises the deficiency of Corollary 5 over departure instants only.

The deficiency oracle is evaluated by mapping every outage sample to its demand cell, forming the (cell $\times$ point) matrix of single-point coverage, taking the longest run per point (`span_util.span_cells`), and then applying the time-window test of Corollary 5. The transport cost (makespan, weighted tardiness, hard-deadline violation) is evaluated exactly at every accepted move, so no Pareto point in Figure 11 is estimated.

Sample size and its consequence. This experiment moves one coordinate of the design space, and the reader should keep that limit in view. The loop perturbs *departure instants only*: batching, routes, aircraft types and sortie-to-vehicle assignments are frozen at the baseline plan, so the neighbourhood explored is the box $[l_i, u_i - d_i]$ around that plan. The search uses 4 restarts of 600 threshold-accepting iterations each, and the deficiency oracle is evaluated at discretised departure instants sampled on the continuous box $[l_i, u_i - d_i]$

Algorithm 3 Co-design feedback loop (threshold accepting on departure instants)

Input: frozen batching, routes, aircraft types and assignments; feasible departure window $[l_i, u_i]$ and duration d_i for each sortie i (so the latest departure that still finishes inside the window is $u_i - d_i$); deficiency oracle $D(\cdot)$; step scale σ

Output: best departure vector and its deficiency

```

1:  $s \leftarrow s^0$ ;  $D^* \leftarrow D(s^0)$ ;  $s^* \leftarrow s^0$ 
2: for  $r = 1$  to  $R$  do ▷ restarts
3:    $s \leftarrow \text{perturb}(s^0)$ 
4:   for  $\text{it} = 1$  to  $I$  do
5:     pick  $i$ ;  $s' \leftarrow s$  with  $s_i \leftarrow \text{clip}(s_i + \mathcal{N}(0, \sigma), l_i, u_i - d_i)$ 
6:      $T \leftarrow T_0 (1 - \text{it}/I)$ 
7:     if  $D(s') \leq D(s)$  or  $\text{rand}() < \exp(-(D(s') - D(s))/T)$  then
8:        $s \leftarrow s'$ 
9:     end if
10:    if  $D(s) < D^*$  then
11:       $D^* \leftarrow D(s)$ ;  $s^* \leftarrow s$ 
12:    end if
13:  end for
14: end for
15: return  $s^*, D^*$ 

```

rather than on the continuous domain itself. The reported response is therefore flat *on that neighbourhood, at that resolution*: the deficiency stays at $23500 \rightarrow 23500$ demand cells under M1 and $1461 \rightarrow 1461$ under M2, with 0 sorties actually moved. A study that re-optimised routes and fleet composition jointly would range over a strictly larger set and could in principle do better; we claim nothing against that. What the null result does support is the narrower statement used in the Discussion: on the frozen baseline plan, retiming departures alone does not close the handover deficiency.

4.6. Controlled terrain family

Because only one digital elevation model is available, we hold the node layout, cargo and parameters fixed and transform the elevation field: relief scaling $z' = z_{\text{med}} + \alpha(z - z_{\text{med}})$, corridor elevation (a 1.8 km-wide corridor whose walls are raised by up to 180 m), and *terrain translation* $z''(i, j) =$

$z'(i + \Delta i, j + \Delta j)$, which changes which obstacle blocks which leg at constant relief. The occlusion ratio – the fraction of sampled transport-track instants without a direct link – is used as the measurable abscissa.

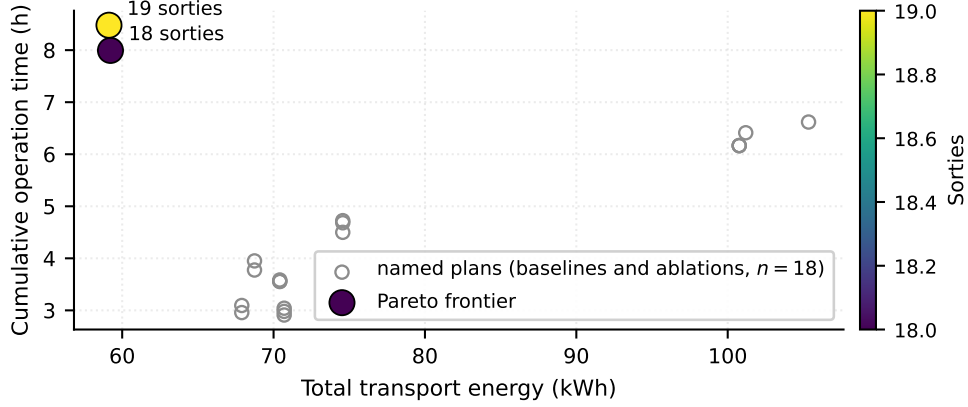


Figure 3: The transport frontier and where the named plans sit. Filled markers are the Pareto frontier of the single-depot routing problem, coloured by sortie count; open circles are the eighteen named plans produced by the baselines and ablations of Section 5.7. Every named plan is dominated on both axes: the frontier trades energy against makespan, and the plans we compare against in the tables fall well inside it.

5. Experiments

5.1. Setup

All experiments share one frozen transport plan and one candidate generation rule, so that only the factor under study varies. The plan comprises 20 sorties carrying 80 boxes, with makespan 12392 s, transport energy 69.093 kWh and on-time rate 83.75%. Every result below is regenerated from the shipped result files.

5.2. Channel-model sensitivity

Table 1 compares the constant-penalty baseline with the ITU-R P. 526 single-edge model and its multi-edge upper bound.

Taking the supplied constant penalty as the reference, moving to knife-edge diffraction *raises* outage demand by +32.2%; taking knife-edge diffraction as

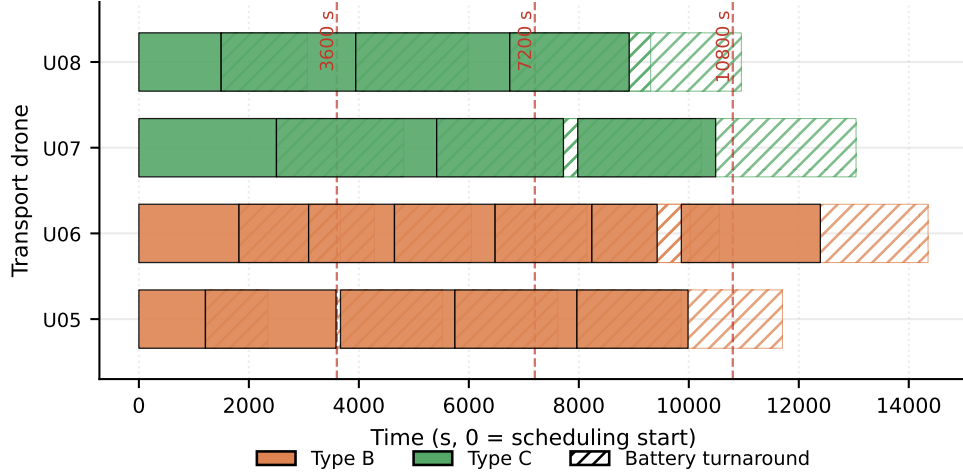


Figure 4: The frozen transport plan, sortie by sortie. Each bar is one sortie on one drone and they do not overlap, so the drone and battery resources are actually respected rather than assumed; the hatched tail of a bar is the battery turnaround that blocks the next sortie on that drone. Dashed vertical lines mark the hard deadlines of the medical and first-batch cargo. The relay layer of Sections 5.3–5.4 is evaluated against exactly this schedule.

Table 1: Channel-model sensitivity (identical transport plan and candidate rule; only the occlusion loss differs).

	constant penalty	P.526 single edge	P.526 upper bound
Outage incidences	1454	1922	1953
Demand cells	992	1052	1055
Usable hover points	2471	1256	1130
Phases	6	5	6
$N = 3$ feasible?	yes	no	no
Smallest N with a verified schedule	3	4	4

the reference instead, the constant penalty admits 49.2% *more* usable hover points, i.e. the P. 526 supply is 49.2% lower. (The two percentages have different bases — 1454 demand instances and 2471 hover points respectively — so they are stated against their own base rather than as one sentence.) Consequently the smallest fleet for which we can construct a verified schedule is not invariant across channel conventions: it is 3 under the supplied constant penalty and 4 under knife-edge diffraction. Two clarifications are needed here,

because the difference matters for how the result may be used.

First, the negative half is a search result, not a proof. At $N = 3$ under P. 526 the constructive verifier reaches cell 236 of 1052 and stops; no recursion and no integer program was run on the P. 526 model. We therefore do not claim that three relays are *infeasible* under P. 526 — only that we did not find a schedule, at any of the three candidate budgets tested (200/400/600 points), whereas the verifier *did* find one at $N = 4$ (7 sorties, no uncovered incidence, SOC minimum 32.1%). The constructive verifier is not a monotone function of N in general — on one realisation of the controlled terrain family (terrain translated by 40 cells) it finds a schedule at $N = 3$ and fails at $N = 4$ — which is a further reason not to read a verifier failure as evidence of infeasibility. On the reference instance it happens to succeed at both $N = 3$ and $N = 4$. The full 4-relay trajectory is shipped, sortie by sortie, with positions, departure and link-completion instants, hover duration, energy and terminal state of charge, so that the positive half can be inspected rather than taken on trust; it is identical at all three candidate budgets tested.

Second, the positive half is what carries the operational message. Whatever the true minimum under P. 526, the verified schedule needs 4 relays where the constant-penalty convention needs 3; the base of the augmentation changes, so a fleet-size claim is a claim about a *particular* occlusion convention and must be reported together with it. Figure 5 shows the two effects separately and then compounded: diffraction raises demand and removes usable hover points, and the smallest constructible fleet is not the same under the two conventions.

5.3. Handover premium

Table 2 separates the coverage baseline from the executable model. Three kinds of statement appear in it, and they must not be conflated: N_{naive} is the *exact* optimum of the coverage-only relaxation (Section 4.3), supplied by an integer program; LB_{handover} is a *proven lower bound* for the handover-aware model, obtained as the minimum number of points covering every union of adjacent phases; and $N^* = 3$ is a *constructed* schedule — an upper bound. Hence $\max(N_{\text{naive}}, LB_{\text{handover}}) = 2 \leq N^* \leq 3$, and the single value 3 should be read as “the smallest fleet for which we have a verified schedule”, never as the minimum.

Why LB_{handover} is a lower bound. The quantity is computed as the largest, over consecutive phase pairs (Φ_i, Φ_{i+1}) , of the minimum number of hover points needed to cover *both* phases’ demand. Since a point count is not

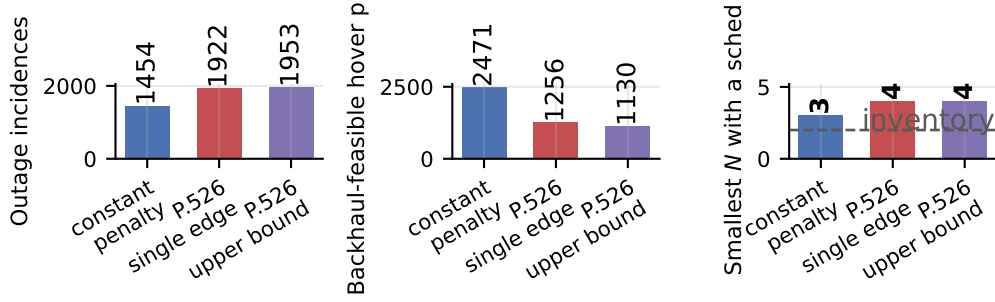


Figure 5: Channel-model sensitivity. Panel (a): outage demand rises by +32.2% once the constant occlusion penalty is replaced by knife-edge diffraction. Panel (b): the supply of backhaul-feasible hover points falls by almost half, because the relay-to-gateway link is diffracted as well. Panel (c): the two effects compound, and the smallest fleet with a constructed schedule rises from 3 to 4; the dashed line marks the two-relay inventory.

Table 2: Relay-side baseline and bounds on the same instance. The status column records whether each entry is an exact optimum, a proven bound, or a constructed schedule.

Quantity	Value	Status
N_{naive} (instant relocation, per-cell set cover)	1	exact (IP)
LB_{handover} (adjacent-phase union cover)	2	proven bound (Prop. 22)
2 (energy bound, under (H))	2	proven bound
Smallest N with a constructed schedule	3	constructed
Handover premium over the relaxation	2	—

obviously a fleet count — one relay may change point between the two phases — we record the argument.

Proposition 22 (Static lower bound from adjacent phases)

Split the mission into phases at the instants where the set of out-of-range sorties changes, and let c_i be the minimum number of hover points whose union covers all demand occurring in phase Φ_i . Then any feasible N -relay schedule satisfies $N \geq \max_i c_i$.

Proof. Fix a feasible schedule and a phase Φ_i . Let P_i be the set of hover points *actually occupied* during Φ_i by the N relays. Condition (V) requires that the union of the points occupied at each instant covers that instant’s demand, and each occupied point is one of the at most N relays. Hence P_i is

a set of at most N points covering all demand in Φ_i , so $|P_i| \geq c_i$ by definition of c_i . Since $|P_i| \leq N$, we get $N \geq c_i$, and i was arbitrary. (Note that this bound does not assume the relays are static across the phase — only that at most N points are occupied at any instant, so a relay that changes point inside a phase only enlarges P_i .) $\square$

Two remarks on the implementation. First, the computed quantity is $\max_i c_i$ over consecutive phase *pairs*, which is the same bound applied to the coarser partition obtained by merging adjacent phases; merging can only lower $\max_i c_i$, so what we report is a valid though looser lower bound, and on this instance it evaluates to 2. Second, the bound is genuinely weaker than the truth: it ignores handover blackout entirely, so it cannot see the $N = 3$ requirement that Theorem 4 and the recursion establish. It is therefore useful as a cheap sanity floor, not as a substitute for the feasibility argument. Figure 6 displays the three quantities together and makes the handover premium visible: the coverage abstraction answers a question nobody is asking, and the executable answer costs 2 more relays.

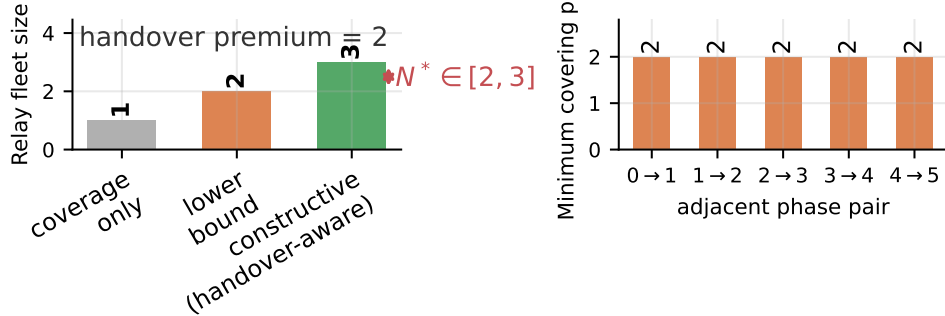


Figure 6: Relay-side baselines. Panel (a): the coverage abstraction that ignores handovers needs $N_{\text{naive}} = 1$ relay, the rigorous lower bound (minimum covering points of every union of adjacent phases) is 2, and the constructive handover-aware verifier finds a schedule with 3. The true optimum is bracketed, and the gap between the abstraction and the executable model — the *handover premium* — is 2 vehicles. Panel (b): the per-phase-pair lower bound is 2 throughout, i.e. no single phase pair can be served by one hovering relay.

Why exactly two relays fail, and on what search space.. The certificate of Corollary 5 is satisfied at 72.0% of the cells under M1, so it does not close the question, and Proposition 15 only asks for 2. The verdict we report

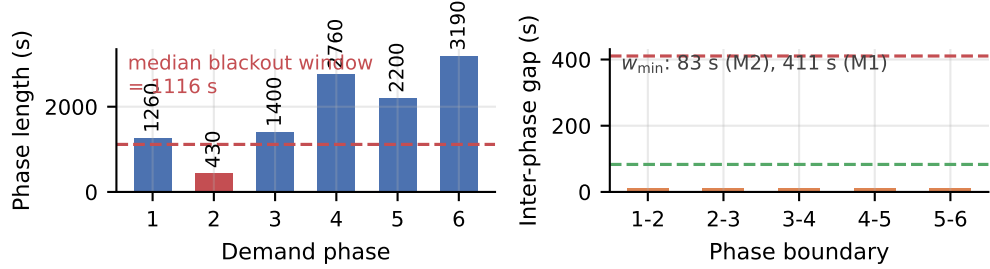


Figure 7: Why relaying cannot keep up, in two scales. Panel (a): the six demand phases are 430–3190s long, and the 430s short phase is shorter than the median blackout window (1116s, dashed) — a single repositioning costs more than that whole phase. Panel (b): the gap between consecutive phases is 10s throughout, against a cheapest repositioning cost of 83s under M2 and 411s under M1. No point change therefore fits into a phase boundary, so any change must occur inside a phase and black out that phase’s demand.

comes from the recursion of Theorem 13, restricted to the candidate set $\mathcal{C}_0$ of points that can individually cover at least one demand cell (2436 of the 2471 hover points) together with the per-cell top-12 refinement: on that set the recursion reports no admissible state at $t = 2420$ s under M1 and $t = 7990$ s under M2, so no two-relay schedule exists whose hover points lie in that set (Proposition 14). The pruning scope of that verdict is stated once in Section 7; an unpruned run at the same breadth was attempted and did not complete within our compute budget, so the evidence class here is “no feasible schedule found by the recursion on a declared finite set”, not an unconditional in-*state-space* decision. What is unconditional is the inner conclusion: *for points drawn from $\mathcal{C}_0 \cup (\text{top-12})$, infeasibility is a statement about a finite, explicitly listed set and does not depend on the order of expansion.*

How the search space itself is chosen. Because the verdict is relative to $\mathcal{C}_0$, we state the rule and the scope explicitly. $\mathcal{C}$ is the dense grid (0.0035° in longitude and latitude, three hover heights above ground) pre-filtered so that both the access and the backhaul link can close at some instant, giving 2471 points. $\mathcal{C}_0 \subseteq \mathcal{C}$ keeps those points that individually cover at least one demand cell (2436 points), and for each cell we additionally keep the top-12 points by remaining coverage. The two nested statements are then:

- **Unconditional in the set:** if the recursion over $\mathcal{C}_0 \cup (\text{top-12})$ with prun-

ing disabled finds no admissible state, no two-relay schedule exists whose hover points lie in that set (Proposition 14). This is a finite, checkable claim.

- **Heuristic in the set:** whether *the instance* has no two-relay schedule depends on whether the optimal hover points lie in $\mathcal{C}_0 \cup (\text{top-12})$. We treat that as a modelling assumption, not a theorem. It is supported by two observations and nothing stronger. First, the relay-side scan of $\mathcal{C}_0$ itself (the relay-side candidate-set scan shipped in **results/**) returns no feasible state at per-cell top-12 or top-24; the top-48 expansion was *not* run to completion, so it is absent from the evidence rather than negative. Second, the reported verdict does not change with the candidate budget in the channel comparison (200/400/600 candidates all give the same smallest fleet), which is weaker evidence, as that budget is the channel experiment’s and not the recursion’s; together they are evidence against — but not a proof of the absence of — a better point outside the set.

The distinction matters because a referee could otherwise object that the recursion is being used as an oracle for a set that was itself chosen heuristically; the items above are the reply.

We also report where the obstruction lives: in the diagnostics the rejected transitions are dominated by “the other relay has not been in place long enough”, i.e. by the alibi requirement, not by the energy budget. Under M1 the handover inequality is satisfiable at the median blackout window for 72.0% of the demand cells under M1 (M2: 93.1%), so a two-relay schedule is forced to freeze its configuration across stretches in which no single configuration covers the demand — exactly what the empty state set detects.

Figure 8 shows the three ingredients together: the span profile $\text{span}(t)$ with its median, the release rate as a function of the blackout threshold, and the resulting bracket on the fleet size.

5.4. Method cross-validation: a phase-level integer program

Because the negative half of the fleet-size answer rests on one algorithm, we solve the same decision question a second time with a structurally different model: a phase-level integer program over the 6 demand phases, with a binary variable per (relay, phase, candidate point), coverage imposed incidence by incidence, and the blackout condition imposed as an incompatibility cut whenever $W(a, b)$ exceeds the interval between two consecutive phases. The model has 480 binaries and 16000 cuts at $N = 2$ and is solved to proven

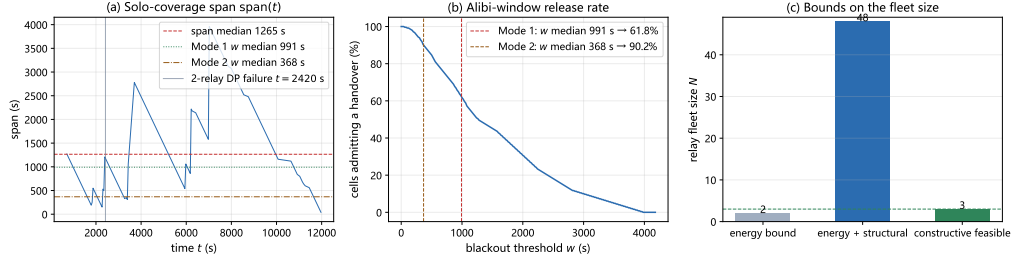


Figure 8: Theory panel. (a) The span profile $\text{span}(t)$ (top 2% of values shaded by the blue curve) together with the median blackout windows of both relocation modes; the gaps in which no single point can hold alone are exactly the stretches in which a repositioning relay cannot be covered. (b) The fraction of demand cells that admit a handover as a function of the assumed blackout threshold: at the median M1 window only 72.0% of cells qualify, against 93.1% for air transit. (c) The proven energy bound of Proposition 15 set against the constructive upper bound; the recursion’s first empty state is *not* a lower bound and is not plotted as one — the interval $2 \leq N^* \leq 3$ is what the instance supports.

optimality in 0.6 s. Table 3 collects the three fleet sizes and places the relaxation’s verdict next to the recursion’s, so that the two structurally different procedures can be compared row by row.

Table 3: Phase-level integer-program relaxation versus the exact cell-level recursion. The third column records whether CP-SAT *closed* the model (status OPTIMAL or INFEASIBLE); note that the shipped result file’s **proven** field uses the narrower meaning “proved infeasible”, so it is **false** for the two satisfiable rows below. A feasible relaxation is *not* a schedule (Proposition 17).

N	Relaxation status	CP-SAT closed?	Implication
1	INFEASIBLE	yes (0.5 s)	$N = 1$ infeasible, independently confirmed
2	OPTIMAL	yes	relaxation satisfiable; <i>not</i> a schedule
3	OPTIMAL	yes	relaxation satisfiable (as expected)

The two models agree on $N = 1$ and disagree on $N = 2$, and the disagreement is the interesting part. The relaxation’s $N = 2$ witness is an explicit assignment that gives each relay one hover point per phase; what it omits is that a relay changing its point *inside* a phase leaves a blackout window that the other relay must cover from a single fixed point. The relevant comparison is therefore with the *gap between* consecutive phases, which is exactly the quantity the relaxation’s condition (ii) constrains: here every one of the 6 – 1

gaps is 10 s wide, whereas the shortest possible repositioning costs $w_{\min} = 83$ s under M2 and 411 s under M1. No point change can therefore be squeezed into a phase boundary at all, so any change must happen inside a phase and black out that phase’s demand. The speculatively relevant alternative — phase *lengths*, which here range from 430 s to 3190 s — is not the binding quantity, and is quoted only to show that the gap, not the length, is what a relocation must fit into. The span profile (1265 s median, 3970 s maximum) is likewise comparable to the blackout window (1116 s median) rather than to a phase. The bottleneck is therefore not the phase-granular resource count — which the relaxation shows is comfortably satisfied at $N = 2$ — but the intra-phase temporal coupling that only the cell-level recursion can see. This is the sharpest form of the answer to “is two relays enough”: *satisfiable at phase granularity, no schedule found at instant granularity*, and the gap is quantified by the release rates of Figure 8.

The relaxation is solved on a reduced candidate set (the 40 points per phase that cover the most incidences), so its *positive* answers carry that restriction, exactly as the recursion’s do; its negative answer for $N = 1$ is strong for the same reason and agrees with both other methods ($N_{\text{naive}} = 1$ coverage abstraction is necessary but not sufficient; the constructive verifier fails at $t = 1980$ s).

5.5. Relocation mode: depot return versus air transit

A natural objection is that infeasibility is an artefact of the depot-return assumption: if a relay could fly directly from one hover point to another, the blackout window would shrink and two relays might suffice. We therefore implemented and compared both modes under the same candidate set and the same energy budget.

The improvement is genuine in magnitude but does not change the conclusion: Figure 10 shows the two effects, and with two relays the obstruction survives both relocation models, so the result is not a consequence of the depot-return assumption. Table 4 gives the verdict fleet size by fleet size.

Three readings of Table 4 matter for the reviewer’s objection. First, the objection is answered: $N = 2$ fails under *both* relocation models, and under M2 the recursion survives to $t = 7990$ s — 3.3 times further in *elapsed time* (cell index 684 versus 171, a ratio of 4.0) than the $t = 2420$ s reached under M1 — before the state set empties, so air transit does buy progress, just not enough. Second, at $N = 3$ the two modes diverge in our search: M1 is certified by an explicit zero-miss schedule, while M2 is not certified. Third,

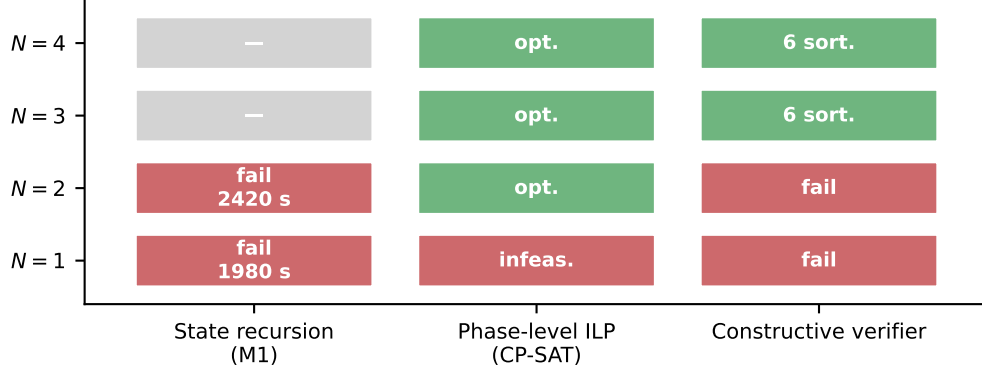


Figure 9: Three independent decision procedures on the same instance. Each cell is the verdict of one procedure at one fleet size: the state recursion of Theorem 13 (M1), the phase-level integer program of Section 5.4, and the constructive verifier of Section 4.3. The three agree on $N = 1$ (no schedule) and, between the two procedures that were run, on the existence of a schedule at $N = 3$ and $N = 4$; they disagree only at $N = 2$, where the relaxation is satisfiable while the instant-level recursion finds no state. That disagreement is the relaxation gap of Proposition 17, and it is why the relaxation is reported alongside the recursion rather than instead of it. The recursion was not run for $N \geq 3$; those cells are left blank rather than filled with an assumption.

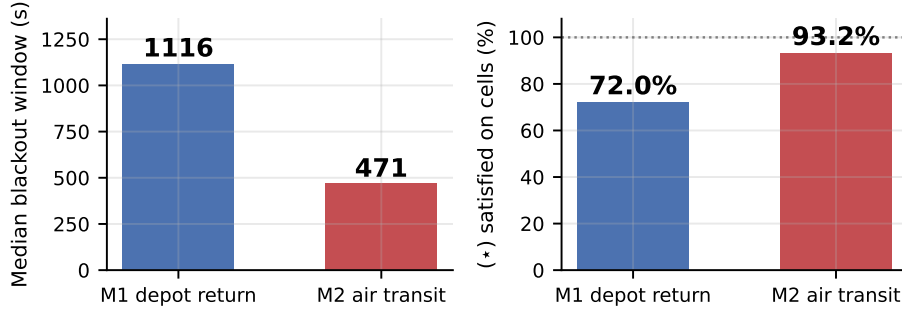


Figure 10: Relocation modes. Panel (a): air transit cuts the median blackout window from 1116s to 471s. Panel (b): the necessary condition of Theorem 4 is satisfied on 72.0% of the demand cells under M1 and on 93.1% under M2 — a real improvement, but the condition still fails somewhere, and no $N = 2$ schedule was found under *either* mode (pruned recursion over the declared set).

Table 4: Dual-mode feasibility verdict. The source column distinguishes the two procedures: *recursion* is the state recursion of Theorem 13 on the declared candidate set with a per-cell top-12 refinement (run with its reported pruning policy, not unpruned), and *verifier* is the constructive height-greedy procedure. A negative verdict from either is a search failure over its scope, not a proof of infeasibility, and the two procedures are not interchangeable: the verifier is not monotone in N in general (on a translated-terrain realisation of the controlled family it finds a schedule at $N = 3$ and fails at $N = 4$), so its failures carry less weight than the recursion’s.

N	Mode M1 (depot return)	Mode M2 (air transit)	Source
1	no state, $t = 1980$ s	no state, $t = 1980$ s	recursion
2	no state, $t = 2420$ s (cell 171)	no state, $t = 7990$ s (cell 684)	recursion
3	feasible: 6 sorties, 0/1454 uncovered, SOC min 33.8%	no schedule found: reached cell 915 of 992 at $t = 11060$ s, after 13 relocations	verifier

that divergence must be read carefully. The M2 verifier made 13 relocations before stalling at $t = 11060$ s, against eight in the M1 run that succeeded: the air-transit heuristic spends its energy budget on more, shorter hops, and the schedule is then cut short by the airborne-time cap rather than by geometry. We therefore report M2 at $N = 3$ as “no schedule found by the verifier”, not as infeasible, and we do *not* claim an air-transit fleet saving: the honest statement is that air transit reduces the blackout window by a factor 1116/471 without changing the $N = 2$ verdict, so the conclusion that the partition needs three relays is robust to Assumption 6’s depot-return clause.

5.6. Is decoupling the cause?

Table 5 reports the co-design loop, in which only the departure instants of the transport sorties are varied; batching, routes, aircraft types and assignments are frozen. The deficiency of Corollary 5 was unchanged by that search: the vector returned at the end of the search is the baseline vector itself (zero sorties shifted, and the trace contains no strictly improving move). Note that the acceptance test is threshold-accepting, so equal-deficiency shifts may be walked through during the search; what is reported is that none of them survives as the returned incumbent — so on this frozen plan, searching over departure instants alone did *not* reduce the obstruction. We state this as what it is: a null result on one coordinate of the design space. Because routing, batching, aircraft assignment and fleet composition were all held fixed, it

is evidence against a purely scheduling-driven explanation of the residual deficiency, not a proof that decoupling is harmless, and not a statement about the joint design space.

Table 5: Co-design feedback loop over departure instants only. The deficiency is counted in demand cells; demand cells are not uniformly spaced in time, so no seconds conversion is reported.

Mode	deficiency before	after	reduction	zero?
M1 (depot return)	23500	23500	0.0%	no
M2 (air transit)	1461	1461	0.0%	no

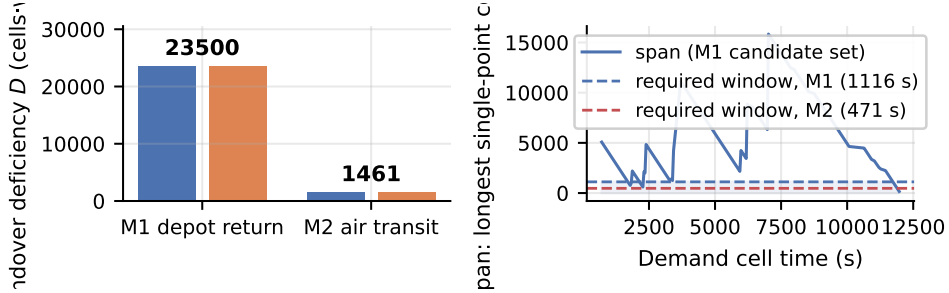


Figure 11: Co-design feedback loop. Panel (a): the handover deficiency is identical before and after re-optimising the departure instants, and the returned vector is the baseline vector (no shift survives as the incumbent), so on this frozen plan the departure instants are not where the residual deficiency can be removed. Panel (b): the span profile — the longest interval over which one single hover point can carry the whole demand — repeatedly falls below the required blackout window (dashed lines), which is the mechanism behind the obstruction.

5.7. Baselines and ablations

Table 7 collects the transport-side method comparison and the partitioning cost. Two caveats apply to that table. The method comparison runs under a fixed iteration budget that differs from the budget used for the frozen plan of Section 5, so its sortie counts are comparable across *methods* but not against the frozen plan; and the rows drawn from the two weight sets (sortie-minimising and on-time-first) are not interchangeable, which is why the CP-SAT status row is drawn from the on-time-first set while the two sortie counts are drawn from the sortie-minimising one. On the transport

layer, a pure construction heuristic needs 19 sorties and 100.76 kWh on the sortie-minimising weight set, against 17 sorties and 67.90 kWh for ALNS; the CP-SAT polishing stage returns an OPTIMAL schedule on its neighbourhood. Partitioning the service area into K autonomous groups raises the weighted resource demand from 1.00 ($K = 2$) to 2.40 ($K = 3$), the *price of partitioning*: splitting duplicates per-group peaks and therefore inflates the fleet.

A textbook baseline, and what it gets wrong.. The four transport methods above all belong to our own algorithmic family, which invites the objection that the comparison is self-referential. We therefore added the most widely cited constructive heuristic for this problem class, the Clarke–Wright savings algorithm [15], generalised from endpoint merging to multi-stop tours as $\text{save}(A, B) = t(A) + t(B) - t(A \cup B)$ on sortie durations and applied until no capacity-, volume- and energy-feasible merge remains. It is evaluated with the *same* scheduler and the same metric function as ALNS, so the comparison is apples-to-apples (Table 6).

Table 6: Clarke–Wright savings versus ALNS under identical scheduling and evaluation code, on the sortie-minimising weight set. The savings heuristic dominates the routing-layer objectives and fails the scheduling ones.

Method	Sorties	Energy (kWh)	Tardiness	Hard viol. (s)	On-time
Clarke–Wright	13	61.26	1210339	30111	61.3%
ALNS	20	76.04	431166	0	80.0%

The savings heuristic starts from 80 single-box sorties (80 vehicles) and merges them down to 13 in 67 merges, reaching 61.26 kWh against ALNS’s 76.04 kWh: it wins decisively on the routing layer. It loses just as decisively on the scheduling layer, with 30111s of hard-deadline violation against 0s and an on-time rate of 61.3% against 80.0%. The mechanism is structural rather than incidental: a route-level merge creates large sorties that must be dispatched early, and they then block each other on the shared drone-plus-battery timeline. A solution with 30111s of hard-deadline violation is not refused by our objective — hard deadlines enter as a penalty, not as a hard constraint — but it is charged every second, so the objective that admits it is 0s worse on that term alone. This is the concrete argument for keeping scheduling inside the search loop instead of repairing it afterwards — the

same conclusion that the relay side reaches from the opposite direction (a coverage abstraction that ignores handovers is not executable).

Table 7: Baselines and ablations (all values from the shipped result files).

Layer	Configuration	Outcome
Transport	pure construction	19 sorties, 100.76 kWh
Transport	Clarke–Wright savings [15]	13 sorties, 61.26 kWh, 30111 s hard violation
Transport	ALNS w/o CP-SAT (ablation)	17 sorties, 67.90 kWh
Transport	ALNS + CP-SAT	OPTIMAL on the fixed neighbourhood
Relay	coverage only (instant relocation)	$N_{\text{naive}} = 1$
Relay	handover-aware, constructive	$N^* = 3$
Modes	M1 vs M2 blackout median	1116 s vs 471 s
Modes	M2 ($\star$) rate	93.1% (vs 72.0% for M1)
Relay	channel model	$N^* = 3$ (constant) vs 4 (P.526)
Coupling	frozen schedule vs co-designed	deficiency unchanged
Partitioning	$K = 1/2/3$ weighted demand	1.00 / 1.00 / 2.40

5.8. What actually determines the fleet size

Figure 12 and Table 8 report the controlled terrain family. After correcting a node-elevation inconsistency (attachment elevations belong to the original terrain and must be re-derived once the elevation field changes; otherwise the aircraft is routed below ground and the resulting outages are spurious), the fleet size turns out to be non-monotone in terrain severity and, among the predictors we computed, no more than weakly associated with the temporal fragmentation of the outage demand. Terrain realisations with almost identical occlusion (44.4%, 44.6%, 45.2%) yield $N^* = 3, 3, > 4$ respectively, while a realisation with only 13.1% occlusion yields $N^* > 4$: three scenarios with the same fragmentation count (0 phases shorter than 1400 s) yield $N^* = > 4, 3, 2$. That spread is the honest summary of this experiment — the pattern is non-monotone, and with nine scenarios we cannot rank the candidate predictors.

Figure 12 plots both panels: the non-monotone step function of panel (a), and panel (b), which rules out the obvious explanation that demand volume alone drives the fleet size.

5.9. Price of partitioning

The integrated ($K = 1$) resource peak is a lower bound on the sum of the per-group peaks; the gap quantifies the cost of autonomous operation.

Table 8: Controlled terrain family. $N^* > 4$ denotes “no feasible schedule found for $N \leq 4$ ”, i.e. a censored observation, not a measured value; 5 of the 9 realisations are censored. The correlations are Pearson coefficients with the censored realisations entered at an arbitrary code 5; the last column gives 95% confidence intervals by the Fisher z -transform, all of which span zero.

Metric	Range	ρ with N^*	95% CI
Occlusion ratio	13.1–45.3%	−0.23	[−0.77, +0.52]
Phases shorter than 1400 s	—	+0.47	[−0.28, +0.86]
Outage incidences	—	−0.12	[−0.73, +0.59]

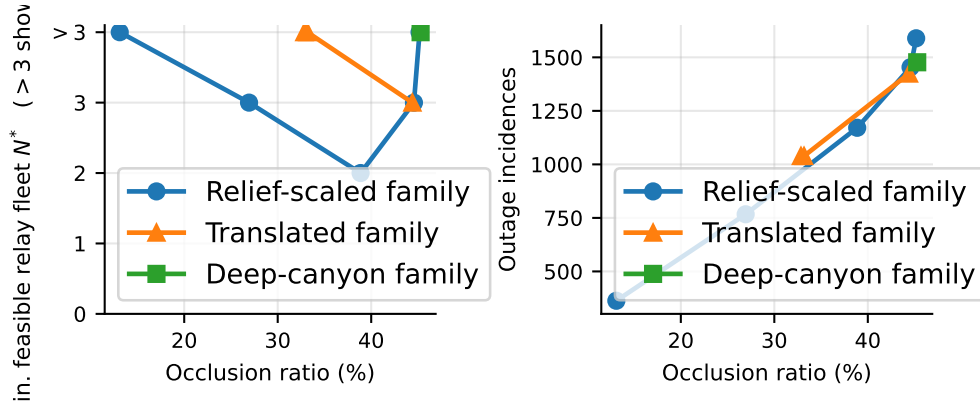


Figure 12: Controlled terrain family (only the DEM elevation field is transformed; nodes and all parameters are frozen, and the node elevations are re-aligned so that no settlement ends up below ground). Panel (a): the smallest fleet with a constructed schedule is *not* monotone in the occlusion ratio — it rises to > 3 at the low end, drops to 3 and then to 2 in the middle, and rises again past 45%. Panel (b): the outage demand itself grows monotonically with the occlusion ratio, so panel (a) is *not* explained by demand volume. Nine scenarios are too few to establish which predictor ranks first; the largest coefficient we computed is for temporal fragmentation (Table 8, $\rho = +0.47$, CI spans zero).

Under the inventory-feasible two-relay plan the per-group relay demand is 2 (integrated) versus 3 under the augmented, continuously-connected plan for $K = 2$; with the channel model of Section 4.4 the same accounting leaves a residual relay gap.

6. Discussion

Relay sizing needs an executable model.. The coverage abstraction answers “how many relays must be in the air at the worst instant”; the operator needs “how many relays must exist”, and the two differ by the handover premium (2 vehicles here). Theorem 4 gives a checkable necessary condition that can be evaluated from the demand profile and the blackout function alone, without solving the scheduling problem; it is not a sufficient certificate, and we quantify how far it is from closing the question (72.0% of the demand cells under M1). Theorem 13 states that the recursion *would be* exhaustive over a stated candidate set with pruning disabled; the runs we report use a per-cell top-12 refinement, so the two-relay verdict is a statement about that declared search space — finite and explicitly listed — rather than about the instance. Proposition 15 supplies an independent, sampling-free lower bound under hypothesis (H), and the constructive verifier a matching upper bound, so the fleet-size interval $2 \leq N^* \leq 3$ is what the instance supports.

The channel convention is part of the answer.. Replacing the constant occlusion penalty with ITU-R P. 526 knife-edge diffraction changes both the demand (+32.2%) and the supply (49.2% of usable hover points), and raises the smallest fleet for which we can construct a verified schedule from 3 to 4. We do *not* claim that three relays are infeasible under P. 526: only a constructive verifier was run there, and it found a schedule at 4 but not at 3. Fleet-size statements are therefore statements about a channel convention *and* a search procedure, and must be reported with both.

Departure instants are not where the obstruction lives.. On the frozen baseline plan, re-optimising departure instants left the obstruction unchanged, so on that plan better delivery scheduling does not close the communication gap; the gap must be managed operationally (pre-programmed autonomous legs, a risk register) or removed by adding relays. Because routing, batching and fleet composition were held fixed, this is a statement about one coordinate of the design space and not about co-design in general.

Occlusion volume is a poor guide, but nine scenarios prove little.. In the terrain study occlusion volume does not order the fleet size, and the non-monotone pattern is real in the sense that realisations with the same fragmentation count take three different values. With 9 scenarios, of which 5

are censored at $N^* > 4$, none of the candidate predictors shows an association whose confidence interval excludes zero, so we present this as a caution against extrapolating fleet size from a single “blocked fraction” rather than as a ranking of predictors.

7. Reproducibility and limits of the reported evidence

What is generated, and what is quoted.. Every table entry and every number in the prose is written by `code/make_numbers_en.py` into a macro file (`numbers_en.tex`, 210 macros), so that no reported value is typed by hand; each macro is traced to its source file in the audit file the script emits. The exclusions are, in full: the correlation coefficients and confidence intervals of Table 8 and the terrain occlusion percentages, computed in the terrain report; the mechanism parameters ($\Delta = 10$ s, the 1400 s short-phase cut-off, the per-cell top-12 depth, the 200/400/600 candidate budgets) together with directly-read summary statistics such as the phase-length range 430–3190 s; the 95% level and censoring code 5 of Table 8; and a few once-reported values (the unpruned 1501 s/555 MB budget, the $K = 1$ baseline of Table 7, the 3.3 time ratio, the relocation counts 8 and 13). Each of these is quoted verbatim from the result file named beside it.

Every other figure and table value is regenerated from the shipped result files by the pipeline below, and each macro is traced to its source file in `numbers_en_sources.txt`. Those file names are Chinese; we keep non-ASCII characters out of the manuscript source for portability.

Three evidence classes.. We use three labels throughout, and they are not interchangeable:

- **Proven:** a mathematical statement with a proof in the text. These are Theorem 4, Corollary 5, Theorem 11, Theorem 18, Proposition 17, Proposition 15 (conditional on hypothesis (H)), Proposition 22, Proposition 2, Proposition 19, Proposition 20, Proposition 21, Proposition 14, Proposition 16, and the lower bound $LB_{\text{handover}} = 2$, which is the exact optimum of a set-cover integer program over unions of adjacent phase pairs.
- **Constructed:** an explicit, independently re-verified schedule. The $N = 3$ relay plan and the transport plan are in this class; we re-check coverage,

timing, energy and state of charge with two independent code paths before reporting them.

- **Search scope:** “no feasible schedule was found” by a specified procedure on a specified finite set. Every negative fleet-size verdict in this paper is in this class — including $N = 2$. In particular the recursion’s verdict is over the declared candidate set C_0 with a per-cell top-12 refinement and the pruning policy reported in Section 7; the run reported here is the pruned one, and the unpruned run is discussed below. We therefore never write “proven impossible” for a negative fleet-size verdict.

What we did not manage to establish.. We record these explicitly so that the scope of the claims is unambiguous. (i) We could not decide the two-relay question over the full candidate set, and we could not even re-run it with state pruning disabled at the smaller top-12 breadth: an unpruned attempt consumed 1501s of CPU (peak 555 MB) without terminating and was stopped by hand, so the verdict stands as a *pruned* recursion over C_0 with the top-12 refinement. That negative result is itself shipped as a result file, with the exact invocation, so a reader can reproduce the non-termination rather than take our word for it. (ii) We could not turn the span inequality into a search-free certificate. Made parameter-free — $W(a, b) \leq \text{span}_{\min}(k)$ with no tuned threshold — it holds for at least one pair in 880 of 992 cells under M1 and 986 of 992 under M2, so it cannot certify infeasibility. (iii) Under the P. 526 channel model no exact decision procedure was run at all; only the constructive verifier was, and it is non-monotone in N . (iv) The terrain study rests on 9 scenarios with 5 censored, which is too few to rank predictors, and the “phase transition” reading is descriptive rather than tested for a bifurcation. (v) The demand profile is evaluated on a fixed sampling grid, so outage statistics are grid-verified rather than proved in continuous time; Corollary 12 shows the qualitative conclusion survives the continuous limit, but the counts do not.

8. Conclusion

We recast hovering-relay feasibility as a handover-aware scheduling problem, proved a checkable necessary condition and a span-based obstruction, replaced a constant occlusion penalty by ITU-R P. 526 knife-edge diffraction, ran a departure-instant-only feedback loop, and gave a nine-scenario sensi-

tivity study of how the fleet size varies across a controlled terrain family. Limitations are stated explicitly and are not incidental. The two-relay verdict, like every other negative fleet-size verdict here, is a statement about a declared finite search space — the candidate set C_0 with a per-cell top-12 refinement, evaluated with the pruning policy we report — and never “proven impossible”; closing state pruning at that breadth did not terminate within our budget, so the verdict is relative to that declared space. The energy bound (Proposition 15) is sampling-free but conditional on hypothesis (H), and Proposition 16 and the span inequality are necessary conditions, not certificates. Under the P. 526 model only a constructive verifier was run, so we claim a constructed schedule at 4 relays and not infeasibility at three. For $N \geq 3$ the verifier is a sufficient procedure, so its failures are search failures. The transport layer is solved heuristically, so no optimality gap is claimed: the reported transport figures are the best feasible solutions found. The terrain family is derived from a single elevation model by controlled transformations, and the outage statistics are verified on a fixed sampling grid rather than in continuous time. Finally, the reference instance uses the elevations supplied with the benchmark; the terrain-family study re-derives them from the transformed elevation field.

Declarations

Data availability. The instance (node coordinates, cargo manifest, fleet and radio parameters) is the benchmark shipped with the problem statement. The 30 m digital elevation model is Copernicus GLO-30 (ESA), openly available. All derived result files, the figure and table generators, and a one-command reproduction pipeline are archived at <https://github.com/ccx111145/data>. Every numerical value in this manuscript is regenerated from those files by `code/make_numbers_en.py`; the bibliography was checked entry by entry against its publisher record by DOI.

CRedit authorship contribution statement. C. Cao: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. Q. Li: Methodology, Software, Validation, Writing – review & editing. Z. Liu: Conceptualization, Methodology, Formal analysis, Supervision, Project administration. Y. Peng: Investigation, Data curation, Visualization. Z. Ouyang: Investigation, Resources, Data curation, Writing – review & editing, Visualization.

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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